Studi Kasus

1.
```
void main() {
double celsius = 30;
double fahrenheit, kelvin;
fahrenheit = (celsius * 9/5) + 32;
kelvin = celsius + 273.15;    print('$celsius°C = $fahrenheit°F');  print('$celsius°C = $kelvin K');}
```

2.
```
void main() {  // Deklarasi variabel
int angka1 = 10;
int angka2 = 5;  // Operasi aritmatika
int penjumlahan = angka1 + angka2;
int pengurangan = angka1 - angka2;
int perkalian = angka1 * angka2;  double pembagian = angka1 / angka2;  int modulo = angka1 % angka2;  // Menampilkan hasil
print('$angka1 + $angka2 = $penjumlahan');
print('$angka1 - $angka2 = $pengurangan');
print('$angka1 * $angka2 = $perkalian');
print('$angka1 / $angka2 = $pembagian');
print('$angka1 % $angka2 = $modulo');
}
```

3.
```
void main() {
String username = 'budi123';
String password = 'pass123';
int umur = 17;
bool isUsernameValid = username.length >= 6;
bool isPasswordValid = password.length >= 6;
bool isAdult = umur >= 18;
bool canRegister = isUsernameValid && isPasswordValid;  bool canAccessAdultContent = canRegister && isAdult;
print('Dapat mendaftar: $canRegister');
print('Dapat mengakses konten: $canAccessAdultContent');}
```

4.
```
void main() {
double hargaBarang = 120000;
int jumlahBeli = 3;
String kodeMember = 'GOLD';
double total = hargaBarang * jumlahBeli;
double diskonPersen = 0;  if (kodeMember == 'GOLD') diskonPersen = 0.1;  else if (kodeMember == 'SILVER') diskonPersen = 0.05;
double diskonTambahan = total > 300000 ? 0.05 : 0;
double totalDiskon = total * (diskonPersen + diskonTambahan);
double hargaAkhir = total - totalDiskon;
print('Total: Rp$total');
print('Diskon: Rp$totalDiskon');
print('Akhir: Rp$hargaAkhir');}
```

5.

```
void main() {

  Map<String, int> nilaiMahasiswa = {
    'Matematika': 85,
    'Fisika': 75,
    'Pemrograman': 90,
    'Bahasa Inggris': 80,
  };

  double total = 0;

  nilaiMahasiswa.forEach((matkul, nilai) {
    total += nilai;
  });

  double rataRata = total / nilaiMahasiswa.length;

  String status = rataRata >= 60 ? 'LULUS' : 'TIDAK LULUS';

  String predikat;

  if (rataRata >= 90) {
    predikat = 'A';
  } else if (rataRata >= 80) {
    predikat = 'B';
  } else if (rataRata >= 70) {
    predikat = 'C';
  } else {
    predikat = 'D/E';
  }

  print("Nilai Mahasiswa:");
  nilaiMahasiswa.forEach((matkul, nilai) {
    print("$matkul: $nilai");
  });

  print("Rata-rata: ${rataRata.toStringAsFixed(1)}");
  print("Status: $status");
  print("Predikat: $predikat");
}
```

```
PROBLEMS 4   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

PS C:\Users\Ahmad Rafi\MOBILE-PROGRAMMING\jobsheet2\dart_application_1> dart run bin/studi_kasus_1.dart
30.0°C = 86.0°F
30.0°C = 303.15 K
PS C:\Users\Ahmad Rafi\MOBILE-PROGRAMMING\jobsheet2\dart_application_1> dart run bin/studi_kasus_2.dart
10 + 5 = 15
10 - 5 = 5
10 * 5 = 50
10 / 5 = 2.0
10 % 5 = 0
PS C:\Users\Ahmad Rafi\MOBILE-PROGRAMMING\jobsheet2\dart_application_1> dart run bin/studi_kasus_3.dart
Dapat mendaftar: true
Dapat mengakses konten: false
PS C:\Users\Ahmad Rafi\MOBILE-PROGRAMMING\jobsheet2\dart_application_1> dart run bin/studi_kasus_4.dart
Total: Rp360000.0
Diskon: Rp54000.00000000001
Akhir: Rp306000.0
PS C:\Users\Ahmad Rafi\MOBILE-PROGRAMMING\jobsheet2\dart_application_1> dart run bin/studi_kasus_5.dart
PS C:\Users\Ahmad Rafi\MOBILE-PROGRAMMING\jobsheet2\dart_application_1>
```

```
PS C:\Users\Ahmad Rafi\MOBILE-PROGRAMMING\jobsheet2\dart_application_1> dart run bin/studi_kasus_5.dart
Nilai Mahasiswa:
Matematika: 85
Fisika: 75
Pemrograman: 90
Bahasa Inggris: 80
Rata-rata: 82.5
Status: LULUS
Predikat: B
PS C:\Users\Ahmad Rafi\MOBILE-PROGRAMMING\jobsheet2\dart_application_1>
```

Praktikum

1.

```
void main() {
  print("=== PROGRAM BMI ===");

  // Deklarasi variabel
  double berat = 70;     // dalam kg
  double tinggi = 1.75;  // dalam meter

  // Validasi sederhana
  if (berat <= 0 || tinggi <= 0) {
    print("Data tidak valid!");
    return;
  }

  // Perhitungan BMI
  double bmi = berat / (tinggi * tinggi);

  // Menentukan kategori
  String kategori;
  if (bmi < 18.5) {
    kategori = "Kurus";
  } else if (bmi < 25) {
    kategori = "Normal";
  } else if (bmi < 30) {
    kategori = "Overweight";
  } else {
    kategori = "Obesitas";
  }

  // Output
  print("Berat: $berat kg");
  print("Tinggi: $tinggi m");
  print("BMI: ${bmi.toStringAsFixed(2)}");
  print("Kategori: $kategori");
}
```

2.

```
/// Program Konversi Mata Uang (Tanpa Input User)

void main() {
  print("=== KONVERSI MATA UANG ===");

  // Jumlah Rupiah
  double rupiah = 150000;

  // Nilai tukar tetap
  const double usd = 15500;
  const double eur = 17000;
  const double jpy = 110;

  // Perhitungan
  double hasilUsd = rupiah / usd;
  double hasilEur = rupiah / eur;
  double hasilJpy = rupiah / jpy;

  // Output
  print("Rupiah: Rp$rupiah");
  print("USD: ${hasilUsd.toStringAsFixed(2)}");
  print("EUR: ${hasilEur.toStringAsFixed(2)}");
  print("JPY: ${hasilJpy.toStringAsFixed(2)}");
}
```

3.

```
/// Demonstrasi Semua Operator Tanpa Import

void main() {
  print("Program Operator Berjalan");

  int a = 10;
  int b = 4;

  // Operator Aritmatika
  print("a + b = ${a + b}");
  print("a - b = ${a - b}");
  print("a * b = ${a * b}");
  print("a / b = ${a / b}");
  print("a % b = ${a % b}");

  // Operator Relasional
  print("a > b = ${a > b}");
  print("a == b = ${a == b}");

  // Operator Logika
  bool x = true;
  bool y = false;
  print("x && y = ${x && y}");
  print("x || y = ${x || y}");
  print("!x = ${!x}");

  // Operator Assignment
  int c = 5;
  c += 3;
  print("c setelah += 3 = $c");

  // Operator Ternary
  String hasil = a > b ? "a lebih besar" : "b lebih besar";
  print("Ternary: $hasil");

  // Null-aware operator
  String? nama;
  print("Nama: ${nama ?? "Tamu"}");

  // Type test
  print("a is int: ${a is int}");
}
```

```
PS C:\Users\Ahmad Rafi\MOBILE-PROGRAMMING\jobsheet2\dart_application_1>  dart run bin/BMI.dart
=== PROGRAM BMI ===
Berat: 70.0 kg
Tinggi: 1.75 m
BMI: 22.86
Kategori: Normal
PS C:\Users\Ahmad Rafi\MOBILE-PROGRAMMING\jobsheet2\dart_application_1> dart run bin/konversi.dart
=== KONVERSI MATA UANG ===
Rupiah: Rp150000.0
USD: 9.68
EUR: 8.82
JPY: 1363.64
PS C:\Users\Ahmad Rafi\MOBILE-PROGRAMMING\jobsheet2\dart_application_1> dart run bin/operator.dart
Program Operator Berjalan
a + b = 14
a - b = 6
a * b = 40
a / b = 2.5
a % b = 2
a > b = true
a == b = false
x && y = false
x || y = true
!x = false
c setelah += 3 = 8
Ternary: a lebih besar
Nama: Tamu
a is int: true
PS C:\Users\Ahmad Rafi\MOBILE-PROGRAMMING\jobsheet2\dart_application_1>
```